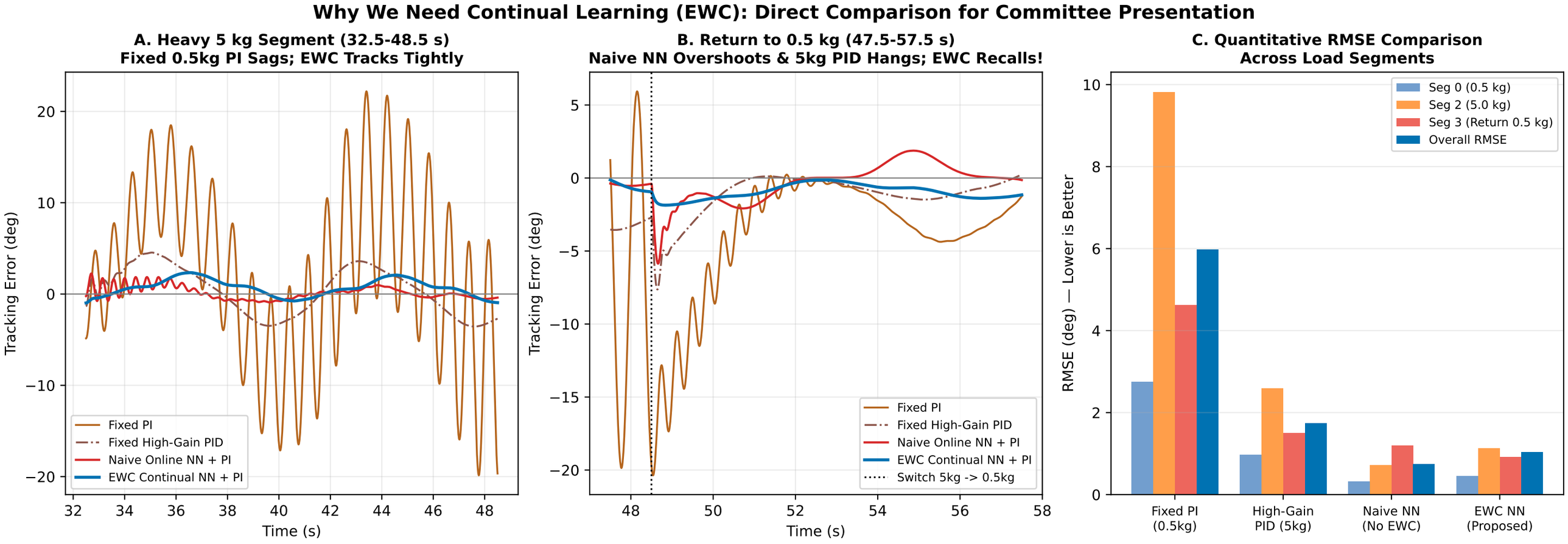

Why We Need Continual Learning (EWC): Direct Comparison for Committee Presentation
A. Heavy 5 kg Segment (32.5-48.5 s)
Fixed 0.5kg PI Sags; EWC Tracks Tightly
Tracking Error (deg)
Time (s)
Fixed PI
Fixed High-Gain PID
Naive Online NN + PI
EWC Continual NN + PI
B. Return to 0.5 kg (47.5-57.5 s)
Naive NN Overshoots & 5kg PID Hangs; EWC Recalls!
Tracking Error (deg)
Time (s)
Fixed PI
Fixed High-Gain PID
Naive Online NN + PI
EWC Continual NN + PI
Switch 5kg -> 0.5kg
C. Quantitative RMSE Comparison
Across Load Segments
RMSE (deg) — Lower is Better
Seg 0 (0.5 kg)
Seg 2 (5.0 kg)
Seg 3 (Return 0.5 kg)
Overall RMSE
Fixed PI (0.5kg)
High-Gain PID (5kg)
Naive NN (No EWC)
EWC NN (Proposed)